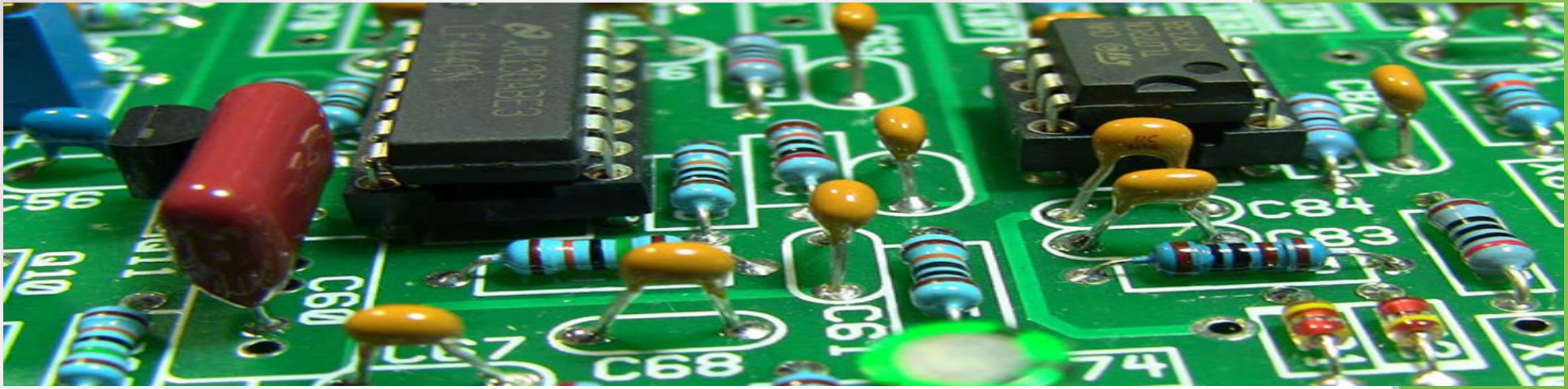
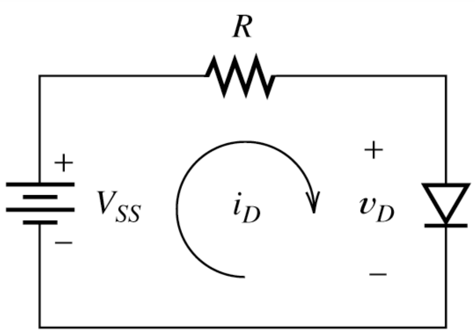


62739
Electromagnetic sensors and digital signal processing
Bipolar transistor

Technical University of Denmark

Bipolar transistor

- **Small base current can control a much larger collector-emitter current.**
- **The base-emitter path can be seen as a diode**



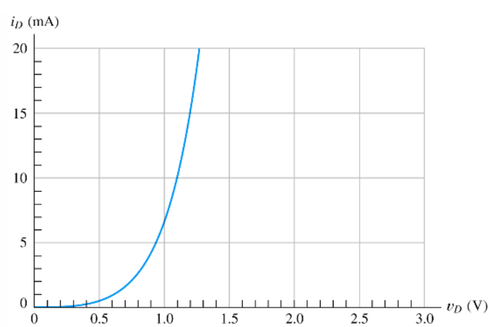


Figure 10.8 Diode characteristic for Exercise 10.3.

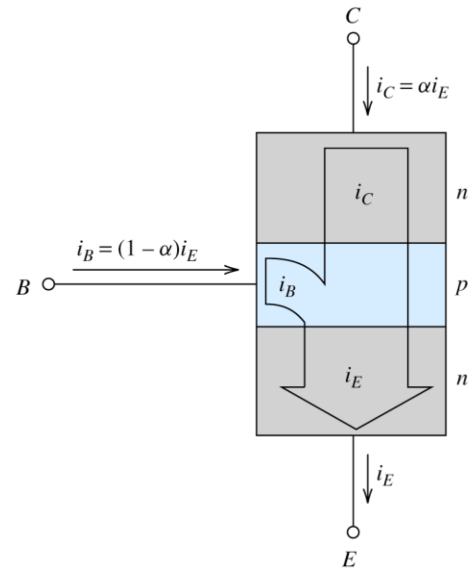


Figure 13.3 Only a small fraction of the emitter current flows into the base (provided that the collector–base junction is reverse biased and the base–emitter junction is forward biased).

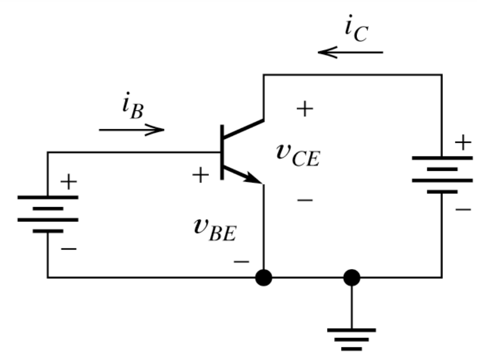
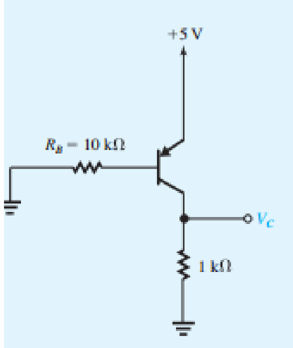


Figure 13.4 Common-emitter circuit configuration for the *npn* BJT.

Bipolar transistor

- **To types NPN and PNP**



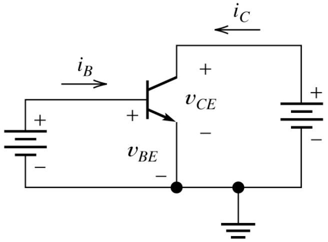


Figure 13.4 Common-emitter circuit configuration for the *npn* BJT.

- **BE and collector characteristics**

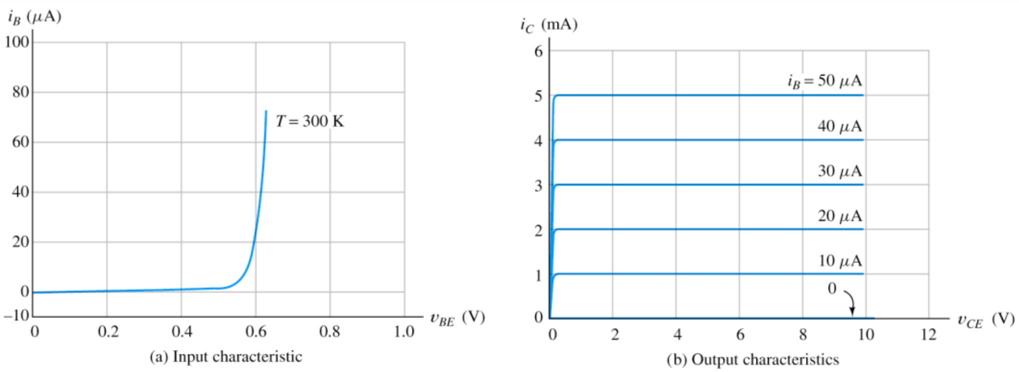
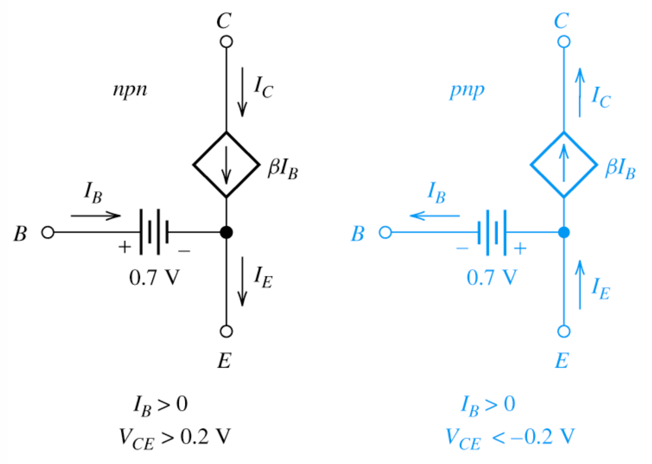


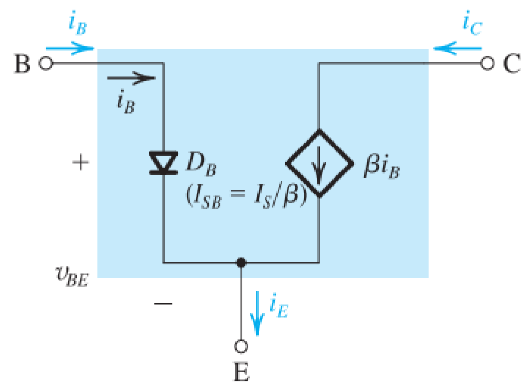
Figure 13.5 Common-emitter characteristics of a typical *npn* BJT.

Bipolar transistor

- **LARGE-SIGNAL DC CIRCUIT MODELS (Active-Region Model)**

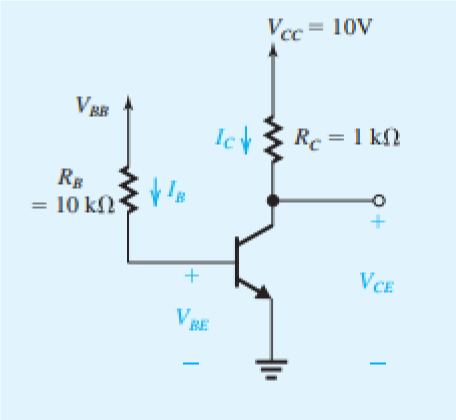


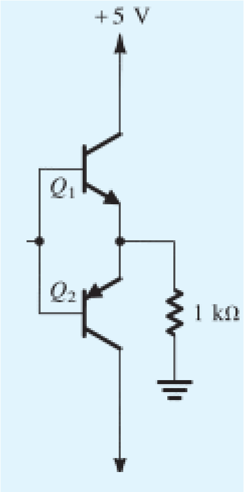
(a) Active region



Bipolar transistor

- **Some common used configurations…**





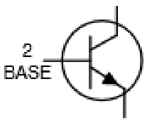
Bipolar transistor

- **Datasheet**
- **BC 337 / 327**

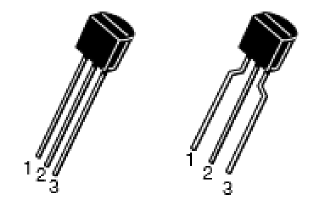
Features

- These are Pb−Free Devices

http://onsemi.com



TO−92
CASE 29
STYLE 17



STRAIGHT LEAD
BULK PACK

BENT LEAD
TAPE & REEL
AMMO PACK

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Collector − Emitter Voltage	V_{CEO}	45	Vdc
Collector − Base Voltage	V_{CBO}	50	Vdc
Emitter − Base Voltage	V_{EBO}	5.0	Vdc
Collector Current − Continuous	I_C	800	mAdc
Total Device Dissipation @ T_A = 25°C Derate above 25°C	P_D	625 5.0	mW mW/°C
Total Device Dissipation @ T_C = 25°C Derate above 25°C	P_D	1.5 12	W mW/°C
Operating and Storage Junction Temperature Range	T_J, T_{stg}	−55 to +150	°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction−to−Ambient	$R_{\theta JA}$	200	°C/W
Thermal Resistance, Junction−to−Case	$R_{\theta JC}$	83.3	°C/W

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

MARKING DIAGRAM

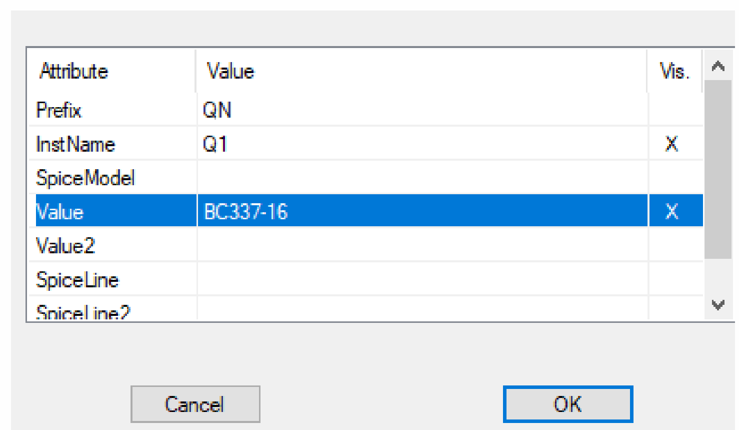
ON CHARACTERISTICS

DC Current Gain (I_C = 100 mA, V_{CE} = 1.0 V) BC337 BC337−25 BC337−40 (I_C = 300 mA, V_{CE} = 1.0 V)	h_{FE}	100 160 250 60	− − − −	630 400 630 −	−
Base−Emitter On Voltage (I_C = 300 mA, V_{CE} = 1.0 V)	$V_{BE(on)}$	−	−	1.2	Vdc
Collector−Emitter Saturation Voltage (I_C = 500 mA, I_B = 50 mA)	$V_{CE(sat)}$	−	−	0.7	Vdc

Adding transistors to LT-spice

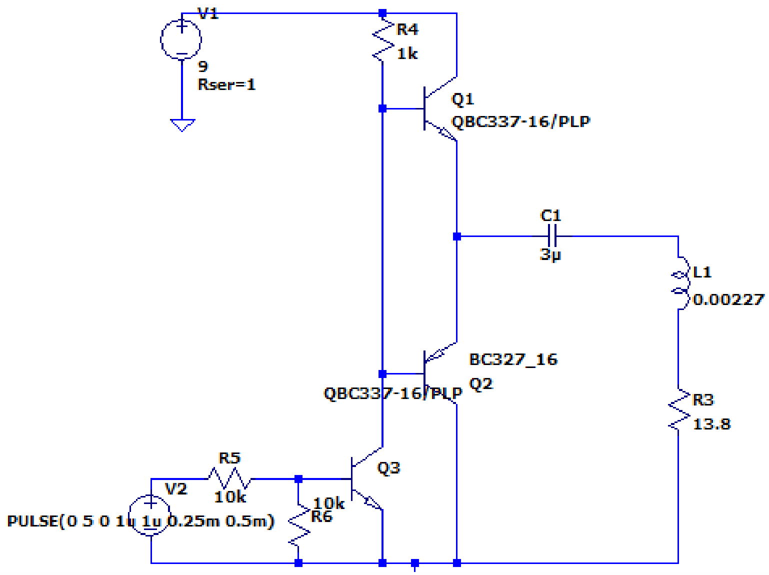
- **Place the models in the user … directory:**
  - EX.: C:\Users\pclyn\Documents\LTspiceXVII\lib\sub
  - Click .op and enter:
  - .lib bc327.sub
  - Add a general NPN transistor to the schematics
  - Open the spice model in an editor and note the component name:
  - CTRL RIGHT click on the NPN component and enter this name in the Value field.

```
.MODEL BC337-16 NPN(IS=4.887E-14 ISE=2.552E-15 ISC=7.006E-14 XTI=3
+ BF=207.2 BR=21.85 IKF=0.902 IKR=0.1 XTB=1.5
+ VAF=184.8 VAR=20 VJE=0.6596 VJC=0.1419
+ RE=0.119 RC=0.25 RB=50 RBM=2 IRB=0.0002
+ CJE=4.217E-11 CJC=1.734E-11 XCJC=0.455 FC=0.652
+ NF=1.002 NR=1.002 NE=1.65 NC=1.25 MJE=0.3434 MJC=0.3484
+ TF=6E-10 TR=3.5E-08 PTF=86 ITF=0.79 VTF=2.4 XTF=2.1
+ EG=1.11 KF=1E-9 AF=1
+ VCEO=45 ICRATING=200M MFG=PHILIPS)
```



Bipolar transistor

- **P.A. stage**



- **Red is base Q1 and Q2 voltages**
- **Blue is current in the coil**